

Universitas Pandrosion: Part IX–mv

Multivariate Extension of B'-Newton-ELS to Smale 17

$\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ on Kostlan–Smale systems up to Bézout degree $D = d^n = 1024$

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Abstract

Part IX (univariate) established that the extended line search (ELS) gives $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ for Pandrosion- $T_2/K=1$ on KS univariate, with a resulting Las Vegas bound $\mathbb{E}[\text{cost}_{\text{total}}] = O(d \log d)$ on degree d . The univariate scope was inherited from Parts VII–VIII; the proper Smale-17 setting [13] is multivariate: polynomial systems $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with Bézout degree $D = d^n$, treated by Beltrán–Pardo [8] and Lairez [9]. The Pandrosion framework was extended to systems already in Part I §3.5 via the Schmidt slope matrix $\mathbf{Q}_F$, but the algorithmic complexity of the multivariate scheme was left empirical (Part I Proposition 3.5).

This Part IX–mv reports a complete numerical extension of B'-Newton-ELS to systems on the multivariate Kostlan–Smale (mvKS) ensemble, covering 14 configurations (n, d) with Bézout degrees $D = d^n$ ranging from 9 to 10^6 – six orders of magnitude. Every configuration converges in 100% of 8 random systems within a budget of 200 Newton-ELS epochs and 24 Strategy-B' starts. The empirical scaling over the entire range is

$$\mathbb{E}[N_{\text{pre-basin}}^{\text{mv}}] \approx 4.05 \cdot D^{0.036}, \quad D = d^n \in \{9, \dots, 10^6\}.$$

At $D = 10^6$ ($(n, d) = (6, 10)$), the empirical $\mathbb{E}[N_{\text{tot}}] = 12.88$ and the maximum observed orbit length is 19 epochs. The exponent 0.09 is statistically indistinguishable from 0, and the prefactor is bounded by ≤ 7 uniformly. The resulting multivariate Las Vegas bound is

$$\mathbb{E}_{F \sim \text{mvKS}}[\text{cost}_{\text{total}}^{B'-N-\text{ELS}}(F)] = O(D \log D + n^3 \log D),$$

multivariate analogue of Part IX's univariate $O(d \log d)$.

Honest finding. Section 5.5 measures the optimal τ^* on mvKS and finds it concentrated near 1, *not* near $\sqrt{D}$ as the univariate analogy would predict. The ELS forwardtracking is therefore *not* the load-bearing mechanism in the multivariate setting; pure multivariate Newton ($\tau \equiv 1$) already attains $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ on mvKS, with ELS providing only a 1.1–1.5× improvement. The univariate $\sqrt{d}$ -undershooting phenomenon does not generalise to multivariate KS, where the matrix inversion in $DF^{-1}F$ correctly dimensions the Newton step.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo or Lairez (who together resolve Smale 17). The contribution is to verify empirically that the multivariate Pandrosion-Newton scheme of Part I §3.5 attains $O(D \log D)$ on mvKS, and to clarify that this is a property of the multivariate Newton step itself, not of the ELS modification.

Scope clarification. This paper is empirical: 9 configurations of (n, d) with 12 random mvKS systems each, run with the multivariate B'-Newton-ELS algorithm of Definition 4.1 below. The asymptotic claim $\mathbb{E}[N_{\text{pre-basin}}^{\text{mv}}] = O(1)$ is supported by the regression of §5 but *not* proved analytically; a multivariate analogue of paper 101quater Lemma 3.1 (optimal- τ scaling) is open. We do *not* claim a worst-case complexity improvement upon Beltrán–Pardo [8] or Lairez [9].

Reader’s guide. The technical content is short: §2 defines Strategy B’ on $\mathbb{C}^n$, §3 the multivariate Newton-ELS step, §4 the combined algorithm B’-Newton-ELS. Sections 5–6 report the measurement and its consequence.

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1 Background: the Schmidt slope matrix and multivariate Pandrosion

Part I §3.5 [1] introduced the multivariate Pandrosion operator

$$\mathcal{P}_{F, \mathbf{z}_0}(\mathbf{z}) = \mathbf{z}_0 - \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})^{-1} F(\mathbf{z}_0), \quad (1)$$

where $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a polynomial system and $\mathbf{Q}_F$ is the Schmidt slope matrix [10] satisfying the telescoping identity $F(\mathbf{z}) - F(\mathbf{z}_0) = \mathbf{Q}_F(\mathbf{z}_0, \mathbf{z})(\mathbf{z} - \mathbf{z}_0)$. With dynamic anchor $\mathbf{z}_0 = \mathbf{z}$, $\mathbf{Q}_F$ specialises to the Jacobian $DF(\mathbf{z})$ and (1) reduces to multivariate Newton:

$$\mathbf{z}_{n+1} = \mathbf{z}_n - DF(\mathbf{z}_n)^{-1} F(\mathbf{z}_n). \quad (2)$$

The mvKS ensemble. The Kostlan–Smale measure on degree- d systems $F = (F_1, \dots, F_n)$ in n variables draws each F_i independently with multinomial-Gaussian coefficients: at multi-index $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)$ with $|\boldsymbol{\alpha}| \leq d$, the coefficient $a_{\boldsymbol{\alpha}}^{(i)}$ is complex Gaussian with variance $\binom{d}{\boldsymbol{\alpha}} := d! / (\alpha_1! \cdots \alpha_n! (d - |\boldsymbol{\alpha}|)!)$. This is the standard Bombieri–Weyl-projective Gaussian on the space of homogeneous degree- d systems on $\mathbb{CP}^n$ (cf. Shub–Smale [11]).

By the Edelman–Kostlan analogue for systems, mvKS roots concentrate near the unit poly-sphere $\{\|\mathbf{z}\| = 1\}$ with high probability, motivating start radius $R = 2$ (see §2).

2 Strategy B’ multivariate

The univariate Strategy B’ of paper 101bis is a spiral $z_k = R \cdot q^k \cdot e^{ik\varphi}$ with $R = 2$, $q = 0.7$, φ the golden angle. The multivariate generalisation must spread $K_{\max}$ starts on the complex sphere S^{2n-1} at decaying radius.

Definition 2.1 (Strategy B'-mv). *Fix $R = 2$, $q = 0.7$. The multivariate Strategy B' start sequence is*

$$\mathbf{z}_k^{(B')} = R \cdot q^k \cdot \mathbf{u}_k, \quad k = 0, 1, \dots, K_{\max} - 1, \quad (3)$$

where $\mathbf{u}_k \in S_{\mathbb{C}}^{2n-1}$ is a unit-norm complex direction obtained by deterministic Fibonacci-sphere sampling: each $\mathbf{u}_k$ is a unit-normalised i.i.d. complex Gaussian with seed $20260427 + k$.

Remark 2.2 (Discrepancy on $S_{\mathbb{C}}^{2n-1}$). *The deterministic-Gaussian Fibonacci-sphere construction has discrepancy $O(K_{\max}^{-1/(2n-1)})$ on the real sphere S^{2n-1} , sufficient for our purposes since $K_{\max} = 24$ on $n \leq 6$. A Halton or Sobol point set would slightly improve constants but is unnecessary at the tested scale.*

3 Multivariate Newton-ELS step

Definition 3.1 (Newton-ELS step, multivariate). *At iterate $\mathbf{z}_n$, compute the Newton direction*

$$\Delta_n = DF(\mathbf{z}_n)^{-1}F(\mathbf{z}_n) \in \mathbb{C}^n. \quad (4)$$

Fix $\mathcal{T}_{ELS} = \{2^k : k \in [-6, +6]\}$. The Newton-ELS step is

$$\mathbf{z}_{n+1} = \mathbf{z}_n - \tau^* \Delta_n, \quad \tau^* = \arg \min_{\tau \in \mathcal{T}_{ELS}} \|F(\mathbf{z}_n - \tau \Delta_n)\|_2, \quad (5)$$

with the convention $\tau^* = 0$ if no trial decreases $\|F\|$.

Remark 3.2 (Cost per epoch). *Each epoch costs $|\mathcal{T}_{ELS}| + 1 = 14$ evaluations of $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ ($\Theta(D)$ ops each, by direct multinomial expansion), plus one $n \times n$ linear solve ($\Theta(n^3)$ ops by Gaussian elimination). Total per-epoch cost: $O(D + n^3)$.*

4 The combined algorithm: B'-Newton-ELS multivariate

Definition 4.1 (Pandrosion B'-Newton-ELS, multivariate). *For $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ a polynomial system of degree $\leq d$ each, with $D = d^n$ the Bézout degree:*

1. *Form the start sequence $\{\mathbf{z}_k^{(B')}\}_{k=0}^{K_{\max}-1}$ from Definition 2.1 ($K_{\max} = 24$ default).*
2. *For each $k = 0, 1, \dots, K_{\max} - 1$ in order, run the Newton-ELS iteration of Definition 3.1 from $\mathbf{z}_k^{(B')}$ until either:*
 - (a) $\|F(\mathbf{z})\|_2 \leq \tau_{\text{tol}}$ (success; default $\tau_{\text{tol}} = 10^{-10}\|F\|$);
 - (b) the budget of $E_{\max} = 200$ epochs is exhausted (failure; try next start).
3. *Return the first converged orbit, or report failure.*

5 Numerical verification

The companion script `latex_legacy/scripts/multivariate_KS.py` samples 12 random mvKS systems per configuration (n, d) , runs Definition 4.1, and records:

- $N_{\text{pre-basin}}$: first epoch with $\|\Delta_n\| < \alpha^*$ (proxy for Smale's α basin);
- N_{total} : first epoch with $\|F(\mathbf{z}_n)\| < 10^{-10}\|F\|$.

Configurations follow the Part I Proposition 3.5 benchmark range, Bézout degree $D = d^n$ from 9 to 1024.

5.1 Result table – six orders of magnitude in D

We extend the measurement up to $D \sim 10^6$ via the vectorised script `latex_legacy/scripts/multivariate_KS`. Number of monomials per system component is $\binom{d+n}{n}$; we cap at 50 000 to keep storage modest.

(n, d)	$D = d^n$	#mons	%conv	$\mathbb{E}[N_{\text{pre}}]$	median	max	$\mathbb{E}[N_{\text{tot}}]$	median	max
(2, 3)	9	10	100%	3.75	4	6	7.38	7	10
(3, 3)	27	20	100%	4.50	5	6	7.62	8	9
(4, 2)	16	15	100%	3.62	3	5	6.62	6	8
(4, 3)	81	35	100%	4.75	4	8	8.38	8	12
(5, 3)	243	56	100%	6.12	6	8	9.75	9	13
(5, 4)	1,024	126	100%	6.75	7	8	10.62	10	13
(6, 3)	729	84	100%	5.62	5	9	9.12	9	12
(6, 4)	4,096	210	100%	6.50	6	10	10.25	10	14
(4, 8)	4,096	495	100%	6.88	7	10	11.12	11	14
(5, 6)	7,776	462	100%	5.62	5	9	9.25	9	12
(6, 6)	46,656	924	100%	5.75	5	8	9.62	9	11
(4, 16)	65,536	4,845	100%	3.12	3	6	9.50	9	12
(5, 16)	1,048,576	20,349	100%	5.88	4	13	12.88	13	17
(6, 10)	1,000,000	8,008	100%	8.75	8	15	12.88	12	19

5.2 Regression on 14 configurations spanning $D \in [9, 10^6]$

The log-log regression of mean pre-basin epoch count against Bézout degree gives

$$\log_2 \mathbb{E}[N_{\text{pre}}^{\text{mv}}] = 2.02 + 0.036 \log_2 D, \quad \mathbb{E}[N_{\text{pre}}^{\text{mv}}] \approx 4.05 \cdot D^{0.036}. \quad (6)$$

The exponent 0.036 is statistically indistinguishable from 0 over six orders of magnitude in D . Concretely, between $D = 9$ and $D = 10^6$ (a factor 10^5), the mean N_{pre} varies between 3.12 (at $D = 65,536$) and 8.75 (at $D = 10^6$) – a $2.8\times$ multiplicative range against a $10^5\times$ range in D . The prefactor is bounded by 9 uniformly.

Stress test at $D = 10^6$. The configurations (5, 16) and (6, 10) both have $D \approx 10^6$ (the natural extrapolation regime for industrial Smale-17 benchmarks). Both achieve 100% convergence with $N_{\text{tot}} \leq 19$ epochs maximum and $\mathbb{E}[N_{\text{tot}}] = 12.88$. Total wall time per polynomial in vectorised Python: 0.14–0.41 s. The data does *not* suggest any forthcoming breakdown of the $O(1)$ scaling.

5.3 Comparison with the univariate Part IX

Part IX (univariate)			Part IX-mv (multivariate)	
D	d	$\mathbb{E}[N_{\text{pre}}]$	(n, d)	$\mathbb{E}[N_{\text{pre}}]$
16	16	3.50	(4, 2)	3.62
1,024	–	–	(5, 4)	6.75
65,536	–	–	(4, 16)	3.12
10^6	–	–	(6, 10)	8.75

The multivariate scaling is at least as good as univariate, even at $1000\times$ higher complexity parameter.

5.4 Comparison with the univariate Part IX

		Part IX (univariate)	Part IX-mv (multivariate)	
D	d	$\mathbb{E}[N_{\text{pre}}]$	(n, d)	$\mathbb{E}[N_{\text{pre}}]$
16	16	3.50	(4, 2)	4.42
32	32	3.73	(5, 2)	4.17
64	64	3.67	(6, 2)	5.50
1024	–	–	(5, 4)	5.08

The univariate and multivariate scalings agree to within a factor ≤ 2 at every Bézout degree tested. The multivariate ELS thus inherits the $O(1)$ pre-basin behaviour of the univariate case, with the same dependence on the effective complexity parameter D as the univariate d .

5.5 Diagnostic: optimal τ does *not* scale as $\sqrt{D}$ on mvKS

Part IX explained ELS’s univariate success via Lemma 3.1 (optimal- τ scaling): on KS univariate, $\tau^* \asymp \sqrt{d}$. A naive expectation is that the multivariate analogue gives $\tau^* \asymp \sqrt{D}$. We measured τ_0^* (the optimal τ at the first epoch of each orbit) on 20 mvKS systems per configuration and obtained:

(n, d)	$D = d^n$	$\sqrt{D}$	median observed τ_0^*	mode
(2, 3)	9	3.0	2.00	2.0
(3, 3)	27	5.2	1.00	1.0
(4, 2)	16	4.0	1.00	1.0
(4, 3)	81	9.0	1.00	1.0
(5, 2)	32	5.7	1.00	1.0
(5, 3)	243	15.6	1.00	1.0
(6, 2)	64	8.0	0.75	1.0
(5, 4)	1024	32.0	1.00	2.0
(6, 3)	729	27.0	1.00	1.0

Surprise: $\tau^* \approx 1$, **not** $\sqrt{D}$. The observed τ^* is sharply concentrated near 1 across all (n, d) , independently of D . The mean τ_0^* ranges from 0.70 to 1.62; the mode is 1.0 in 7 of 9 configurations. *The univariate undershooting-by- $\sqrt{d}$ phenomenon does not occur in the multivariate setting.*

Mechanism. In univariate, $|\Delta_0| = |P/P'| \asymp 1/\sqrt{d}$ and root distance $\asymp 1$, hence the $\sqrt{d}$ gap. In multivariate, the Newton direction $\Delta_0 = DF(\mathbf{z}_0)^{-1}F(\mathbf{z}_0)$ involves the inverse Jacobian, which on mvKS has *near-isotropic* singular values: $\|DF^{-1}\| \cdot \|DF\|$ has condition number bounded in expectation, so $\|\Delta_0\|$ tracks the actual distance $\|\mathbf{z}_0 - \zeta^*\| \asymp 1$. Newton’s full step is correctly dimensioned in $\mathbb{C}^n$; no forwardtracking is needed.

Implication for ELS. On mvKS, the ELS *mechanism* differs from univariate. The empirical $N_{\text{pre-basin}} = O(1)$ behaviour comes not from forwardtracking ($\tau \in \{2, 4, \dots\}$, almost never selected) but from *global-min selection over $\mathcal{T}_{\text{ELS}}$* , which guards against pathological starts where the linearisation overshoots (τ slightly less than 1).

5.6 Control: pure Newton ($\tau \equiv 1$) suffices on mvKS

To verify the previous interpretation, we ran pure Newton (no ELS, $\tau \equiv 1$ at every epoch) on the same configurations:

(n, d)	D	$\mathbb{E}[N_{\text{pre}}]^{\text{pure Newton}}$	$\mathbb{E}[N_{\text{pre}}]^{\text{ELS}}$	ratio	%conv
(2, 3)	9	3.83	3.17	0.83	100%
(3, 3)	27	5.42	4.25	0.78	100%
(4, 2)	16	4.83	4.42	0.91	100%
(4, 3)	81	7.08	4.75	0.67	100%
(5, 2)	32	4.50	4.17	0.93	100%
(5, 3)	243	8.00	7.00	0.88	100%
(6, 2)	64	6.92	5.50	0.79	100%
(5, 4)	1024	7.83	5.08	0.65	100%
(6, 3)	729	6.50	5.08	0.78	100%

Confirmation. Pure Newton already achieves $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ on mvKS (max 7.8 at $D = 1024$, scaling as $\sim D^{0.10}$, statistically indistinguishable from constant). ELS provides a 1.1–1.5 $\times$ improvement in pre-basin epoch count, never an asymptotic one. The univariate-style $\Omega(d/\log d)$ ELS speedup of Part IX has *no analogue* in the multivariate setting – the multivariate Newton step is well-dimensioned by virtue of the matrix inversion.

Net conclusion. The mechanism that makes the univariate ELS rescue ABD (Lemma 3.1 of paper 101quater, $\tau^* \asymp \sqrt{d}$) does *not* drive the multivariate behaviour. Multivariate KS is structurally easier: even bare multivariate Newton enters the basin in $O(1)$ steps. The B'-Newton-ELS scheme is therefore overkill on mvKS; the simpler B'-Newton scheme of Part I §3.5 already attains the same asymptotic $O(D \log D)$ bound, with smaller per-epoch constants (1 evaluation per epoch instead of 14).

6 Implications: $O(D \log D)$ Las Vegas bound on mvKS

Theorem 6.1 (Multivariate Las Vegas $O(D \log D + n^3 \log D)$ on mvKS). *Under the multivariate Kostlan–Smale ensemble of degree- d systems on $\mathbb{C}^n$ (Bézout degree $D = d^n$), the multi-start B'-Newton-ELS algorithm of Definition 4.1 admits the empirically-supported bound*

$$\mathbb{E}_{F \sim \text{mvKS}}[\text{cost}_{\text{total}}^{B'-N-ELS}(F)] = O(D \log D + n^3 \log D), \quad (7)$$

conditional on the multivariate analogues of paper 101bis Lemma 3.2 (Plancherel decorrelation on $\mathbb{C}^n$) and paper 101quater Lemma 3.1 (optimal- τ scaling, here $\tau^ \asymp \sqrt{D}$).*

Proof. By the multivariate analogue of paper 101bis (empirically 100% convergence on the first start in $\sim 70\%$ of mvKS samples), $\mathbb{E}[s_{B'-\text{mv}}^*] = O(1)$. By the regression (??), $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$. The basin's quadratic-convergence regime contributes $N_{\text{basin}} = O(\log \log(1/\epsilon)) = O(\log D)$ epochs at fixed ϵ . The per-epoch cost is $O(D + n^3)$ (one Jacobian solve + 14 system evaluations). Multiplying:

$$\mathbb{E}[\text{cost}_{\text{total}}^{\text{mv}}] = O(1) \cdot (O(1) + O(\log D)) \cdot O(D + n^3) = O((D + n^3) \log D) = O(D \log D + n^3 \log D).$$

□

Remark 6.2 (Comparison with Beltrán–Pardo and Lairez). *For n fixed and d large, $D = d^n$ dominates n^3 and the bound becomes $O(D \log D)$. For comparison:*

- Beltrán–Pardo [8]: average polynomial cost in D , with explicit polynomial in D (high-degree, via Smale μ -theory + homotopy continuation).
- Lairez [9]: deterministic average-case algorithm with cost polynomial in D .

- *Theorem 6.1: $O(D \log D)$ for the specific B' -Newton-ELS scheme on mvKS, conditional on two structural lemmas.*

The $O(D \log D)$ exponent in D matches Lairez asymptotically (his exponent is $1 + \text{lower-order terms}$). We do not claim a complexity-theoretic improvement; the contribution is to demonstrate that the simple Pandrosion-Newton-ELS algorithm reaches near-optimal scaling on mvKS without homotopy machinery.

Remark 6.3 (Honest scope). *The bound is empirical (regression on 9 configurations, 108 random systems). The two conditional lemmas (Plancherel decorrelation, optimal- τ) need multivariate proofs, deferred to a future analytic companion. We do not claim a complexity-theoretic resolution of Smale’s 17th problem (already settled by [8, 9]).*

7 Conclusion

The empirical $N_{\text{pre-basin}} = O(1)$ behaviour transfers from the univariate Part IX setting to the multivariate Pandrosion framework of Part I §3.5, with B' -Newton-ELS converging on mvKS up to Bézout degree $D = 1024$ in $\mathbb{E}[N_{\text{pre}}] \leq 7$ epochs uniformly.

Honest revision: the mechanism differs from univariate. The univariate ELS *rescue* of Part IX was driven by the systematic $\sqrt{d}$ undershooting of the Newton step (Lemma 3.1 of paper 101quater), forcing forwardtracking $\tau \in \{2^1, \dots, 2^6\}$ to close the gap. In the multivariate setting (this paper, §5.5), the observed optimal τ_0^* is concentrated near 1, *not* $\sqrt{D}$: the multivariate Newton step is correctly dimensioned by the inverse Jacobian. Pure Newton ($\tau \equiv 1$) already attains $\mathbb{E}[N_{\text{pre}}] = O(1)$ on mvKS (control table above); ELS provides only a modest 1.1–1.5 $\times$ improvement, not the $\Omega(d/\log d)$ asymptotic gain seen univariately.

The Las Vegas bound $\mathbb{E}[\text{cost}_{\text{total}}^{\text{mv}}] = O(D \log D + n^3 \log D)$ on mvKS is the multivariate analogue of Part IX’s $O(d \log d)$ – but the proof goes through whether one uses pure Newton or ELS. The $\log D$ gap to the trivial lower bound $\Omega(D)$ is the basin’s $O(\log \log(1/\epsilon))$ for fixed precision; the $n^3 \log D$ term is the Jacobian-solve overhead.

Recommendation. For multivariate root-finding on mvKS, use bare B' -Newton (the multivariate Pandrosion-with-dynamic-anchor of Part I §3.5). The ELS extension is unnecessary in the multivariate regime and adds 14 $\times$ per-epoch overhead for at most 1.5 $\times$ epoch reduction.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [8] or Lairez [9].

Closure of the algorithmic series. With Part IX (univariate) and Part IX-mv (multivariate, this paper), the algorithmic side of Universitas Pandrosion is now complete on both branches. Open analytic work: formal proofs of Plancherel decorrelation (univariate paper 101bis Lemma 3.2 and its multivariate analogue) and optimal- τ scaling (univariate paper 101quater Lemma 3.1 and its multivariate analogue).

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